

Traveling-Salesman Approximation Exercises

Question–Answer Notes for Exercises 35.2-1 to 35.2-5

Background

Metric TSP Reminder

A complete undirected graph has a **metric** cost function if, for all vertices u, v, w ,

$$c(u, w) \leq c(u, v) + c(v, w).$$

This is the triangle inequality. It is the key assumption behind the standard TSP approximation algorithms.

1 Exercise 35.2-1

Question 35.2-1

Suppose that a complete undirected graph $G = (V, E)$ with at least 3 vertices has a cost function c satisfying the triangle inequality. Prove that

$$c(u, v) \geq 0$$

for all $u, v \in V$.

Answer

Assume for contradiction that some edge has negative cost:

$$c(u, v) < 0.$$

Since $|V| \geq 3$, choose a third vertex w distinct from u and v .

Proof

Using the triangle inequality on the triple u, v, w gives

$$c(u, w) \leq c(u, v) + c(v, w).$$

Since $c(u, v) < 0$, this implies

$$c(u, w) < c(v, w).$$

Now use the triangle inequality in the other direction. Since the graph is undirected, $c(v, u) = c(u, v)$. Thus

$$c(v, w) \leq c(v, u) + c(u, w) = c(u, v) + c(u, w).$$

Again, since $c(u, v) < 0$,

$$c(v, w) < c(u, w).$$

We have obtained both

$$c(u, w) < c(v, w) \quad \text{and} \quad c(v, w) < c(u, w),$$

which is impossible. Therefore no edge can have negative cost, and

$$c(u, v) \geq 0$$

for all $u, v \in V$.

2 Exercise 35.2-2**Question 35.2-2**

Show how in polynomial time we can transform one instance of the traveling-salesman problem into another instance whose cost function satisfies the triangle inequality. The two instances must have the same set of optimal tours. Explain why such a polynomial-time transformation does not contradict the negative approximation result for general TSP, assuming $\mathbf{P} \neq \mathbf{NP}$.

Answer

Assume the original TSP instance has nonnegative edge costs. Let

$$m > \max_{u \neq v} c(u, v).$$

Define a new cost function h by

$$h(u, v) = c(u, v) + m$$

for every edge (u, v) .

Triangle Inequality

For any three vertices u, v, w ,

$$h(u, v) = c(u, v) + m \leq 2m.$$

Since all original costs are nonnegative,

$$h(u, w) + h(w, v) = c(u, w) + m + c(w, v) + m \geq 2m.$$

Therefore

$$h(u, v) \leq h(u, w) + h(w, v),$$

so the new cost function satisfies the triangle inequality.

Same Optimal Tours

Every Hamiltonian tour on n vertices uses exactly n edges. Therefore every tour receives exactly the same additive increase:

$$nm.$$

If a tour had original cost C , its transformed cost is

$$H = C + nm.$$

Since the same constant nm is added to every tour, the ordering of tours by cost does not change. Hence the original and transformed instances have exactly the same set of optimal tours.

Why This Does Not Contradict the Hardness Result

Suppose a ρ -approximation algorithm is run on the transformed metric instance. Let H be the transformed cost returned by the algorithm, and let H^* be the transformed optimum. Then

$$H \leq \rho H^*.$$

Returning to the original costs,

$$C = H - nm \quad \text{and} \quad C^* = H^* - nm.$$

Thus

$$C + nm \leq \rho(C^* + nm),$$

and so

$$C \leq \rho C^* + (\rho - 1)nm.$$

Key Point

The additive term $(\rho - 1)nm$ prevents this from being a constant-factor approximation guarantee for the original instance. Therefore the transformation does not contradict the negative result for approximating general TSP.

3 Exercise 35.2-3

Question 35.2-3

Consider the closest-point heuristic for metric TSP:

- start with a cycle containing one arbitrary vertex;
- repeatedly choose the vertex u outside the cycle that is closest to any vertex already on the cycle;
- if v is the closest vertex on the cycle, insert u just after v .

Prove that this heuristic returns a tour of total cost at most twice the cost of an optimal tour.

Answer

The closest-point choices are exactly the choices made by Prim's algorithm for building a minimum spanning tree. The insertion rule maintains a preorder tour of that tree. Therefore the heuristic is another way to produce the standard MST preorder tour used in the 2-approximation for metric TSP.

Proof

Root the construction at the first vertex r . Whenever the heuristic inserts a new vertex u after the closest cycle vertex v , draw the tree edge (v, u) . Because u is the vertex outside the current set that is closest to the current set, these are exactly Prim edges. Hence the chosen edges form a minimum spanning tree T .

The cyclic order maintained by the heuristic is a preorder order of the tree constructed so far:

- the parent v is already on the cycle before its child u is added;
- inserting u immediately after v places the child directly after its parent in the current preorder order.

By induction, after all vertices are inserted, the final tour is the preorder Hamiltonian tour obtained from T .

Now use the standard metric-TSP analysis. If H^* is an optimal TSP tour, then removing one edge from H^* gives a spanning tree. Since T is an MST,

$$c(T) \leq c(H^*).$$

A full walk around T has cost

$$2c(T) \leq 2c(H^*).$$

The preorder Hamiltonian tour is obtained from this full walk by shortcutting repeated vertices. By the triangle inequality, shortcutting cannot increase the cost. Therefore the heuristic's tour H satisfies

$$c(H) \leq 2c(T) \leq 2c(H^*).$$

Thus the closest-point heuristic is a 2-approximation for metric TSP.

4 Exercise 35.2-4**Question 35.2-4**

In the bottleneck traveling-salesman problem, we want a Hamiltonian cycle that minimizes the cost of the most costly edge in the cycle. Assuming the cost function satisfies the triangle inequality, show that there is a polynomial-time approximation algorithm with approximation ratio 3.

Answer

Compute a bottleneck spanning tree, then convert it into a Hamiltonian cycle whose every edge skips over at most three tree edges. The triangle inequality then gives a factor-3 bound.

Algorithm

1. Compute a bottleneck spanning tree T of the complete graph. This can be done in polynomial time; for example, any minimum spanning tree is also a bottleneck spanning tree.

2. Let

$$B = \max_{e \in T} c(e).$$

3. Take a full walk of T and shortcut it to visit every vertex exactly once, using the tree-cube property: consecutive recorded vertices are at distance at most 3 in T .
4. Return the resulting Hamiltonian cycle H .

Tree-Cube Lemma

For every tree T , the vertices can be ordered cyclically so that consecutive vertices in the order are at tree distance at most 3. Equivalently, the third power of every tree has a Hamiltonian cycle. This ordering can be found recursively from a full walk of the tree by shortcutting repeated vertices without skipping more than two consecutive intermediate tree vertices.

Approximation Proof

By the tree-cube lemma, every edge in the returned Hamiltonian cycle H replaces a path of at most 3 edges in T . Each edge of T has cost at most B . Therefore, by the triangle inequality, every edge of H has cost at most

$$3B.$$

So the bottleneck cost of H satisfies

$$b(H) \leq 3B.$$

Let H^* be an optimal bottleneck Hamiltonian cycle, and let

$$B^* = b(H^*)$$

be the cost of its most expensive edge. If we delete any edge from H^* , the remaining edges form a spanning tree whose maximum edge cost is at most B^* . Since T is a minimum bottleneck spanning tree,

$$B \leq B^*.$$

Therefore

$$b(H) \leq 3B \leq 3B^*.$$

Conclusion

The algorithm returns a Hamiltonian cycle whose bottleneck edge is at most three times the optimum bottleneck edge. Thus metric BOTTLENECK-TSP has a polynomial-time 3-approximation algorithm.

5 Exercise 35.2-5

Question 35.2-5

Suppose the vertices of a TSP instance are points in the plane and the cost $c(u, v)$ is the Euclidean distance between points u and v . Show that an optimal tour never crosses itself.

Answer

If a tour has two crossing edges, we can uncross them and obtain a strictly shorter tour. Therefore a crossing tour cannot be optimal.

Proof

Suppose a tour contains two crossing edges (x_1, x_2) and (y_1, y_2) . Let P be their intersection point. Removing these two edges breaks the tour into two paths. Reconnect the paths using the noncrossing pair of edges, for example (x_1, y_1) and (x_2, y_2) , reversing one of the two paths if necessary.

The new object is still a Hamiltonian tour: it visits the same vertices exactly once, but it uses two new noncrossing edges instead of the two crossing edges.

Now compare the lengths. Since P lies strictly inside both crossing segments, the triangle inequality is strict in the two triangles:

$$|x_1y_1| < |x_1P| + |Py_1|,$$

and

$$|x_2y_2| < |x_2P| + |Py_2|.$$

Adding these inequalities gives

$$|x_1y_1| + |x_2y_2| < |x_1P| + |Px_2| + |y_1P| + |Py_2|.$$

But the right-hand side is exactly

$$|x_1x_2| + |y_1y_2|.$$

Therefore

$$|x_1y_1| + |x_2y_2| < |x_1x_2| + |y_1y_2|.$$

The uncrossed tour has strictly smaller total cost, while all other edges are unchanged. This contradicts optimality of the original tour. Hence an optimal Euclidean TSP tour has no crossings.

Final Summary**What to Remember**

- Triangle inequality plus symmetry forces all edge costs to be nonnegative when there are at least three vertices.
- Adding a large constant to every edge creates a metric instance with the same optimal tours, but it does not preserve approximation ratios for the original instance.
- The closest-point insertion heuristic is Prim's MST preorder tour in disguise, so it is a 2-approximation for metric TSP.
- Metric bottleneck TSP has a 3-approximation via a bottleneck spanning tree and the tree-cube shortcut property.
- Euclidean optimal TSP tours never cross, because crossing edges can be uncrossed to reduce total length.